Financial Time Series

TA Session #5

Friday, April 23, 2021

statsmodels.stats.diagnostic.het_arch(std_resid)

Engle Stat .919 F stat .999

dist of std_resid ~ Normal plot
Q-Q

fit a dist for the $\{\varepsilon_t\}$

forecasting - point prediction using the mean model m_t

- $\hat{\sigma}_t \longrightarrow \hat{\sigma}_{t+h}$

fitted dist of $\{\varepsilon_t\}$ process (Q-Q)

$\hat{\sigma}_{t+k} \mathcal{N}(0,1) + \hat{m}_{t+k}$

compute 99% quantile

could be t_{df}

std_resid = results.residual/results.conditional_volatility

$a_t = \varepsilon_t \sigma_t$ $\quad \frac{\varepsilon_t \sigma_t}{\sigma_t} \Rightarrow \varepsilon_t$

H_0 vs H_A Test Statistic Engle test stat

no ARCH effect F test test stat

$\sigma_t = \text{const}$

AR(p) $\quad X_t = \overset{=0}{\phi_0} + \cancel{\phi_i}\sum_{i=1}^{p} \phi_i X_{t-i} + \varepsilon_t$

$$X_{t-1} = X_{t-1}$$

$$X_{t-2} = X_{t-2}$$

$$\vdots$$

$$X_{t-(p-1)} = X_{t-(p-1)}$$

$$\begin{pmatrix} X_t \\ \vdots \\ X_{t-(p-1)} \end{pmatrix} = \begin{pmatrix} \phi_1 & \cdots & & \phi_p \\ 1 & 0 & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & \cdots & 1 & 0 \end{pmatrix} \begin{pmatrix} X_{t-1} \\ \vdots \\ X_{t-p} \end{pmatrix} + \begin{pmatrix} \varepsilon_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$\underset{\sim}{X}_t = \underset{\sim}{M}\, \underset{\sim}{X}_{t-1} + \underset{\sim}{\varepsilon}_t$$

$$
\begin{aligned}
\underset{\sim}{X}_t &= \underset{\sim}{M}\, \underset{\sim}{X}_{t-1} + \underset{\sim}{\varepsilon}_t \\
&= \underset{\sim}{M}\left(\underset{\sim}{M}\, \underset{\sim}{X}_{t-2} + \underset{\sim}{\varepsilon}_{t-1}\right) + \underset{\sim}{\varepsilon}_t \\
&= \underset{\sim}{M}^2 X_{t-2} + \underset{\sim}{M}\, \varepsilon_{t-1} + \underset{\sim}{\varepsilon}_t \\
&= \cdots \\
&= \underset{\sim}{M}^t X_0 + \sum_{i=0}^{t-1} \underset{\sim}{M}^{(i)} \varepsilon_{t-i}
\end{aligned}
$$

for stationarity need $\underset{\sim}{M}^t \to 0$ as $t \to \infty$

$M = ADA^T$, $M^t = AD^tA^T$ $\quad D^t \to 0$

$$D^t = \begin{pmatrix} \lambda_1^t & & 0 \\ & \ddots & \\ 0 & & \lambda_p^t \end{pmatrix} \qquad |\lambda_i| < 1$$

$$\det(M - \lambda I) = 0$$

$$\begin{pmatrix} \phi_1 - \lambda & \phi_2 & \cdots & \phi_p \\ 1 & -\lambda & & 0 \\ 0 & \ddots & \ddots & \vdots \\ 0 & & 1 & -\lambda \end{pmatrix} = 0$$

$$\begin{pmatrix} \phi_1 - \lambda & \phi_2 \\ 1 & -\lambda \end{pmatrix} = 0 \qquad 0 = (-\lambda)(\phi_1 - \lambda) - \phi_2$$

$$= \lambda^2 - \lambda\phi_1 - \phi_2$$

Roots of $\lambda^2 - \lambda\phi_1 - \phi_2 = 0$ must be inside the unit circle

$$|\lambda_i| < 1$$

Let $z = \frac{1}{\lambda}$ $\qquad \frac{1}{z^2} - \frac{\phi_1}{z} - \phi_2 = 0$

$-\phi_2 z^2 - \phi_1 z + 1 = 0$ autoregressive polynomial

roots must be outside unit circle

This becomes

$$\lambda^{p} - \lambda^{p-1}\phi_1 - \dots - \lambda^{0}\phi_p = 0$$

$$1 - z\phi_1 - \dots - z^{p}\phi_p = 0$$

in the general case

GARCH Forecasting: 7

X

$$\hat{\sigma}^2_{t+k} = \omega + (\alpha_1 + \beta_1)\hat{\sigma}^2_{t+k-1}.$$

We let $\lambda = \alpha_1 + \beta_1$ and iterate backward: $0 < \lambda < 1$

$$\begin{aligned}
\hat{\sigma}^2_{t+k} &= \omega + \lambda \underline{\hat{\sigma}^2_{t+k-1}} \\
&= \omega(1+\lambda) + \lambda^2 \hat{\sigma}^2_{t+k-2} \\
&= \cdots \\
&= \omega\underbrace{(1 + \lambda + \cdots + \lambda^{k-2})}_{\to \frac{\omega}{1-\lambda}} + \lambda^{k-1}\hat{\sigma}^2_{t+1}
\end{aligned}$$

As $k \to \infty$, assuming $\lambda < 1$

$$\hat{\sigma}^2_{t+k} \to \sigma^2 = \omega/(1-\lambda).$$

λ = "persistence"

The half-life of the volatility difference is approximately $\lambda^T = \frac{1}{2}$, so $T \approx -\frac{\log 2}{\log \lambda} = -\frac{.693}{\log \lambda}$